

C2_02 Gate C Protected Review - Astra RC4

Subscription billing / payment reliability. Open only after a saved independent Gate C attempt; then close the review and perform changed repair proof.

1. Objective controls

Control	Expected	Why
Invoices	95	invoice grain
Paid status	57 / 184743.17	invoice-status contract
Payments	83	event grain
Success events	64	event count; not invoice count
Success invoices	63 / 202098.12	invoice amount counted once
Account IDs	18 raw / 17 unique	account_id 8 conflicts
Orphan payment	invoice 9999 / 650	exclude from invoice KPI

2. Expected reasoning path

Start with invoice vs payment-event grain: joining all payment attempts to invoice rows can multiply billed amount. Aggregate payment evidence to invoice grain before invoice-level money KPIs.

Duplicate key: account_id 8 maps to Enterprise/East and Pro/South and is used by five invoices. Plan/region ranking is blocked until that mapping is repaired or the affected rows are quarantined and disclosed.

Two metric contracts: invoice status=paid gives 57 invoices / 184743.17. Has at least one successful payment event gives 63 invoices / 202098.12 when invoice amount is counted once. Neither number is self-evidently the one KPI; name the business contract.

Retry control: multiple failed/success payment events can exist for one invoice. Payment-event counts may count retries; invoice amount must not be summed once per event.

Orphans: invoice 94/account999 is cancelled and payment event 83/invoice9999 is a successful 650 orphan. Keep both visible; never let the 650 enter invoice-paid money.

3. Critical failures

- Uses INNER JOIN to make orphans disappear without an explicit exception population.
- Uses DISTINCT as a generic fan-out repair instead of proving key/cardinality.
- Calculates a KPI at event grain while calling it ticket/order/invoice grain.
- Uses LAG before proving complete/deterministic time grain.
- Ranks a dirty dimension attribute before resolving a conflicting duplicate key.
- Reports an attractive total without independent row/distinct-key/money controls.
- Treats EXPLAIN or fast runtime as proof that business logic is correct.

4. Competency floors

Competency	Required proof
Grain/cardinality	Input/output grain; duplicate and orphan treatment; no unexplained fan-out.
SQL architecture	Readable CTE/subquery stages with controls at the correct grain.
Window/time	Partition/order/frame or LAG contract; deterministic period logic and gap awareness.
Validation	Rows, distinct keys, numerator/denominator and money independently reconcile.
Performance/handoff	Safe EXPLAIN evidence after correctness; query comments/assumptions and rerun path.

Changed repair proof: after review, close this PDF and solve a different grain/key/time case. Do not use the same Gate fixture as repair evidence. Gate C is the final fresh reassessment for this module.